

# Thermal history of Mars inferred from orbital geochemistry of volcanic provinces

David Baratoux<sup>1,2</sup>, Michael J. Toplis<sup>1,2</sup>, Marc Monnereau<sup>1,2</sup> & Olivier Gasnault<sup>1,2</sup>

Reconstruction of the geological history of Mars has been the focus of considerable attention over the past four decades, with important discoveries being made about variations in surface conditions<sup>1</sup>. However, despite a significant increase in the amount of data related to the morphology, mineralogy and chemistry of the martian surface, there is no clear global picture of how magmatism has evolved over time and how these changes relate to the internal workings and thermal evolution of the planet. Here we present geochemical data derived from the Gamma Ray Spectrometer on board NASA's Mars Odyssey spacecraft<sup>2</sup>, focusing on twelve major volcanic provinces of variable age. Our analysis reveals clear trends in composition that are found to be consistent with varying degrees of melting of the martian mantle. There is evidence for thickening of the lithosphere ( $17\text{--}25\text{ km Gyr}^{-1}$ ) associated with a decrease in mantle potential temperature over time ( $30\text{--}40\text{ K Gyr}^{-1}$ ). Our inferred thermal history of Mars, unlike that of the Earth, is consistent with simple models of mantle convection<sup>3–6</sup>.

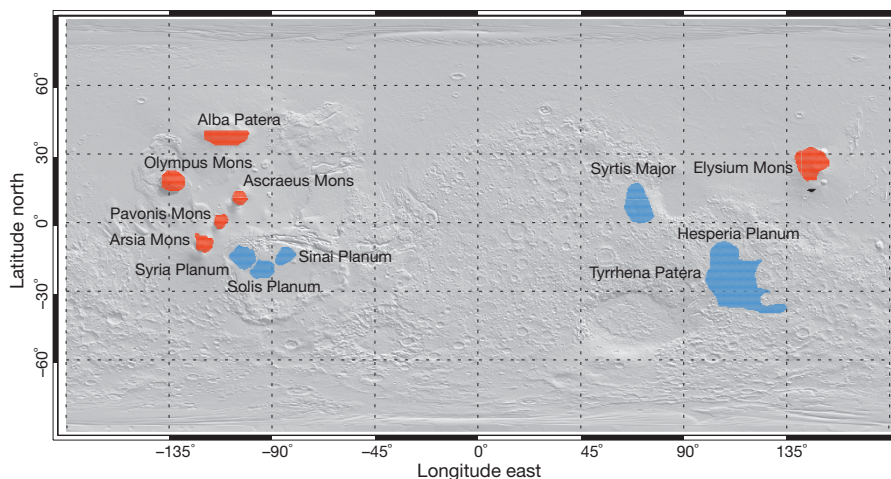
Partial melting of planetary interiors leads to the formation of liquids whose composition is directly related to the pressure–temperature conditions of melting. If such liquids reach the surface, for example as lava flows, this opens the possibility of using surface composition to infer degrees and depths of melting as functions of time. The link between magma chemistry and mantle thermal state has been used to rationalize certain variations in terrestrial magmatism<sup>7</sup>, but has never been applied to other planetary bodies, principally because of a lack of appropriate data.

The Gamma Ray Spectrometer (GRS) instrument provides the first opportunity to explore this possibility elsewhere than on the Earth, as the abundance of the rock-forming elements silicon, iron, potassium, thorium and chlorine has been quantified at the martian surface over an equatorial band extending from  $\sim 45^\circ$  north to  $\sim 45^\circ$  south<sup>2</sup>.

Consideration of the entire GRS data set has been used to identify regions of similar chemistry<sup>8,9</sup>, although detailed interpretation of these provinces is complicated by the fact that surface chemistry may be affected by primary magmatism, subsequent chemical alteration and transport processes. To highlight magmatic signatures, an alternative approach is to restrict analysis to regions with clear morphological evidence for a volcanic origin<sup>10</sup> (for example lava flows, volcanic vents, calderas or shield volcanoes). Regions have been selected that are significantly larger than the GRS footprint and which fall well within the larger units previously defined on the basis of GRS data<sup>9</sup>. As such, average chemical composition is independent of the exact shape and size of each region. With these criteria, we retained 12 provinces (Fig. 1), six of Hesperian age (3.6–4.0 Gyr) and six of more recent, Amazonian, age.

Our analysis focuses on iron, silicon and thorium. We did not explicitly consider potassium as variations of K/Th are remarkably modest<sup>11</sup> and thorium is known to be less mobile than potassium during aqueous alteration. Chlorine was not considered as although it may be of magmatic origin<sup>12</sup>, it is also known to be concentrated in secondary salts<sup>13</sup>. Average and standard deviations of FeO (parts per hundred by weight (wt%)), SiO<sub>2</sub> (wt%) and thorium (parts per million by weight (wt p.p.m.)) in each region were calculated from GRS data released to NASA's Planetary Data System<sup>2</sup> and renormalized on a hydrogen-free basis (Table 1).

All three elements show significant variability and a prominent anticorrelation is observed between SiO<sub>2</sub> and thorium (Fig. 2a), with younger volcanoes being characterized by lower SiO<sub>2</sub> and higher thorium contents (Fig. 2a). Values of FeO are generally lower for the younger edifices of the Tharsis dome (Fig. 2b). Comparison of these trends with possible global dust compositions argues against the possibility that geochemical variations result from a variable fraction of transported material (Supplementary Information).



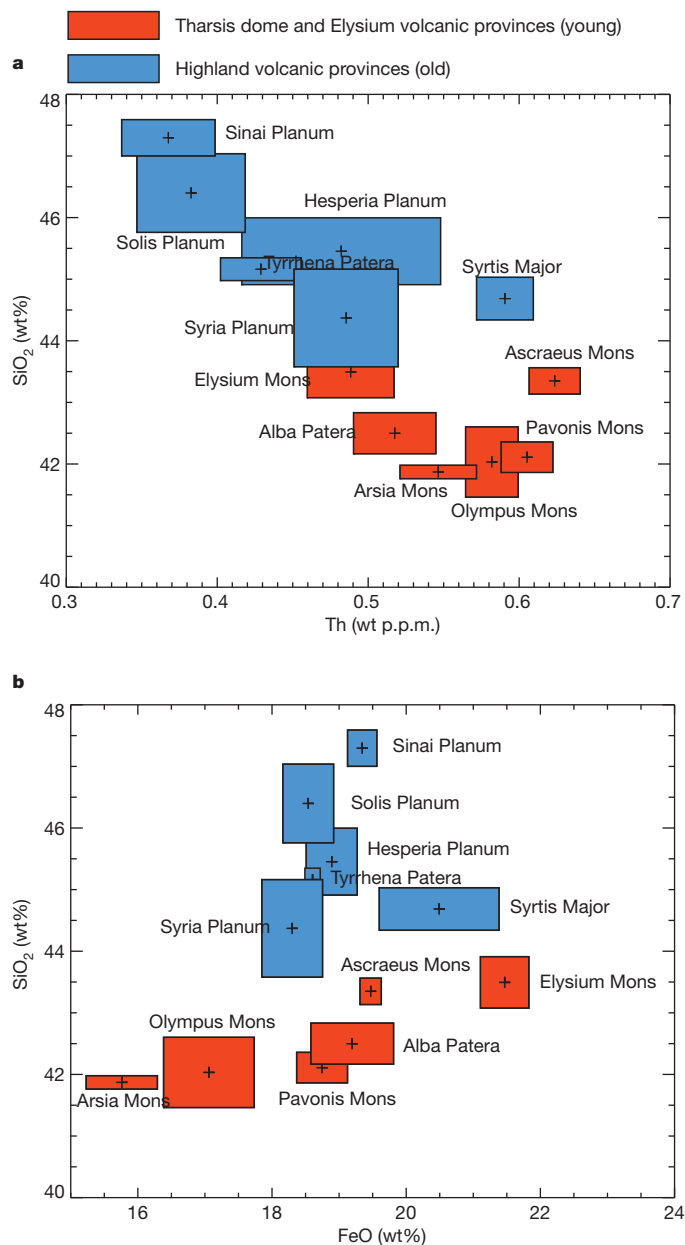
**Figure 1 | Selected volcanic provinces represented over a shaded relief map of Mars.** Hesperian volcanic provinces, located in the highlands south of the hemispheric dichotomy, are highlighted in blue and Amazonian provinces occurring as large shield volcanoes are highlighted in red.

<sup>1</sup>Université de Toulouse, UPS-OMP, IRAP, F-31400 Toulouse, France. <sup>2</sup>CNRS, IRAP, 14 Avenue Edouard Belin, F-31400 Toulouse, France.

**Table 1 | Average composition and s.d. for each volcanic province**

| Volcanic province | SiO <sub>2</sub> (wt%) | FeO (wt%)  | Th (wt p.p.m.) | Age (period) |
|-------------------|------------------------|------------|----------------|--------------|
| Olympus Mons      | 42.0 ± 0.6             | 17.1 ± 0.7 | 0.58 ± 0.02    | Amazonian    |
| Arsia Mons        | 41.9 ± 0.1             | 15.7 ± 0.5 | 0.55 ± 0.03    | Amazonian    |
| Ascræus Mons      | 43.4 ± 0.2             | 19.5 ± 0.2 | 0.62 ± 0.02    | Amazonian    |
| Pavonis Mons      | 42.1 ± 0.2             | 18.7 ± 0.4 | 0.60 ± 0.02    | Amazonian    |
| Alba Patera       | 42.3 ± 0.3             | 19.2 ± 0.6 | 0.51 ± 0.03    | Amazonian    |
| Elysium Mons      | 43.5 ± 0.4             | 21.5 ± 0.3 | 0.49 ± 0.03    | Amazonian    |
| Syrtis Major      | 44.7 ± 0.3             | 20.5 ± 0.9 | 0.58 ± 0.02    | Hesperian    |
| Tyrrhena Patera   | 45.2 ± 0.2             | 18.6 ± 0.1 | 0.43 ± 0.03    | Hesperian    |
| Hesperia Planum   | 45.5 ± 0.5             | 18.9 ± 0.4 | 0.48 ± 0.02    | Hesperian    |
| Syria Planum      | 44.4 ± 0.02            | 18.3 ± 0.5 | 0.58 ± 0.02    | Hesperian    |
| Solis Planum      | 46.4 ± 0.6             | 18.5 ± 0.4 | 0.38 ± 0.04    | Hesperian    |
| Sinai Planum      | 47.3 ± 0.3             | 19.3 ± 0.2 | 0.36 ± 0.03    | Hesperian    |

All values are normalized to water-free composition.



**Figure 2 | Abundances of silica, thorium and iron oxide for the 12 selected areas shown in Fig. 1. a, Th versus SiO<sub>2</sub>. b, FeO versus SiO<sub>2</sub>. Older Hesperian volcanoes are shown in blue and younger volcanic provinces are shown in red. Box sizes correspond to errors in element concentration of 1σ.**

Assuming a magmatic origin, interpretation of these variations requires an estimate of the mantle composition and an understanding of the magmatic processes leading to volcanic eruption. Concerning mantle chemistry, constraints from chondritic and shergottite–nakhilite–chassignite<sup>14</sup> meteorites have been used to estimate the composition of the primitive martian mantle<sup>15</sup>, a composition henceforth referred to as DW85. Although there is some debate surrounding the question of whether or not this composition is the source of shergottites<sup>16,17</sup> the DW85 composition has been shown to be consistent with Tharsis volcanism<sup>10</sup> and, most importantly, with basaltic rocks analysed *in situ* at Gusev crater<sup>18</sup>.

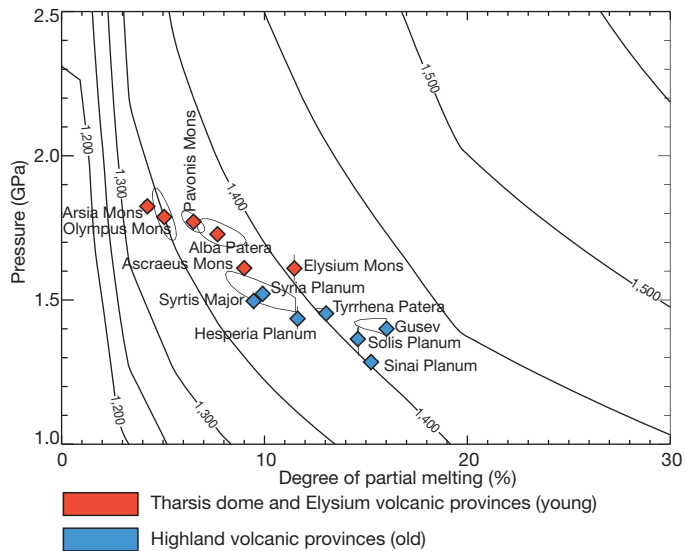
In terms of magmatic processes, we assume that melts are produced by adiabatic decompression of the DW85 composition associated with convective upwelling and that liquids equilibrate with the mantle at a single pressure and temperature at the base of the lithosphere. We begin by considering the simplest case, of no fractional crystallization on the way to the surface. Although primary magmas are rare on the Earth, consideration of magnesium number (Mg#, defined as molar Mg/(Mg + Fe) × 100) suggests that they may be much more common at the surface of Mars<sup>19</sup>. For example, whole-rock data from Gusev crater show that the spatially dominant basalts have Mg# ranging from 51 to 55 (ref. 19), consistent with that calculated for primary melts in equilibrium with the DW85 mantle (49 ± 4). Furthermore, the olivine in Shergottite meteorites has core compositions with Mg# typically of 70 to 80, consistent with the range expected for olivine crystallizing from primary magmas derived from a DW85 source. Although these comparisons support the idea of limited degrees of fractional crystallization, we also note that our derived pressures of melting are insensitive to <15% of olivine removal (Methods and Supplementary Information).

We determined optimal values of pressure (*P*) and the degree of partial melting (*F*) for each volcanic province by comparison with primary melt compositions calculated using the pMELTS software<sup>20</sup> (Methods). Pressure is generally well constrained, but a significant range of *F* may explain the observed silica/iron oxide abundances (Methods and Supplementary Information). However, an independent assessment of *F* is possible using thorium concentrations. Being highly incompatible, the concentration of this element in the liquid is inversely proportional to the degree of partial melting. Assuming the thorium value of DW85 (0.056 wt p.p.m.), values of *F* derived from silica–iron systematics and from thorium concentrations are in excellent agreement for the six Hesperian provinces. This leads to the conclusion that the DW85 mantle composition is relevant to Hesperian volcanism and that a model of direct transfer of magma from the mantle to the surface is sufficient. However, for younger volcanism values of SiO<sub>2</sub>, FeO and thorium do not always point to a consistent *P*–*F* solution.

The effects of low-pressure fractional crystallization, assimilation of cumulates, variations in the Mg# of the mantle and thorium depletion have all been tested (Methods and Supplementary Information). Of these, only the last two can potentially account for the Amazonian data. The concentration of highly incompatible elements in the mantle is sensitive even to small amounts of previous melting (Methods and Supplementary Information). For this reason, the hypothesis of a modest depletion of thorium in the Amazonian mantle relative to the value of DW85 has been retained.

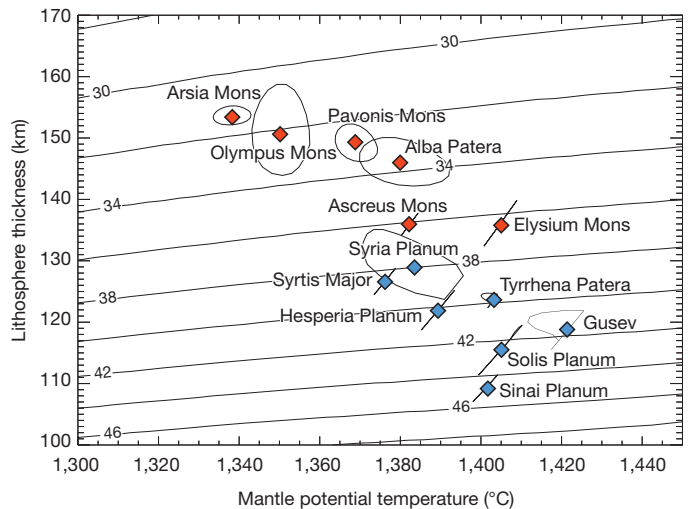
The results (Fig. 3) show that older volcanism is characterized by lower *P* and higher *F* values relative to younger volcanism (typically >10% melting at pressures of <1.5 GPa for older volcanism and <10% melting at pressures of >1.5 GPa for younger volcanism). As an independent check, we used *in situ* measurements of iron and silicon concentrations of the Hesperian-age Adirondack-class basalts in Gusev crater<sup>18,19</sup> to calculate *P* and *F* (Fig. 3). These derived values are consistent with those determined from experimental phase equilibria (15–20% melting at 1.4 GPa (ref. 18)) and are comparable to those of our Hesperian volcanic group.

The variations in *P* and *F* imply a change in the thermal state of the mantle that may be quantified by consideration of potential



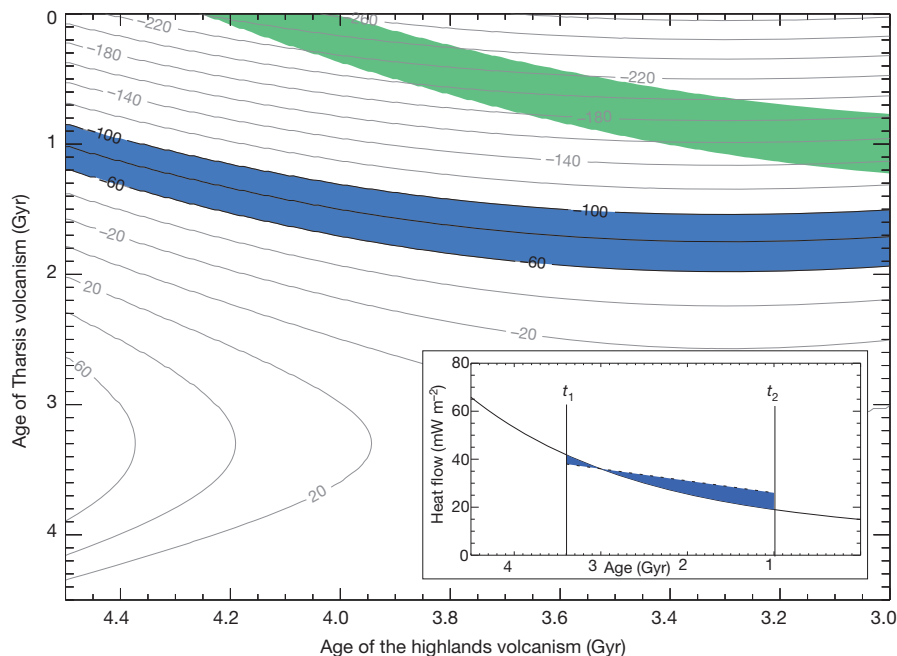
**Figure 3 | Pressure and degree of partial melting for each volcanic province.** Pressures are derived from  $\text{SiO}_2$  and  $\text{FeO}$  abundances (Methods). In general,  $F$  is determined from thorium content in the DW85 mantle except when inconsistent with values derived from  $\text{SiO}_2$  and  $\text{FeO}$ . In these cases,  $F$  values derived from  $\text{SiO}_2$  and  $\text{FeO}$  concentrations are given, retaining the hypothesis of a mantle depleted in thorium (Supplementary Information).  $1\sigma$  error bars and ellipses correspond to propagated uncertainties in  $\text{SiO}_2$  and  $\text{FeO}$ . Contours correspond to mantle potential temperature in degrees Celsius, calculated by adding the latent heat of fusion (equivalent to  $\sim 4$  K for each per cent of melting) to, and subtracting the effect of adiabatic decompression to surface pressure from, the equilibrium temperature at which liquid and solid mantle coexist<sup>21</sup>.

temperature<sup>21</sup> (that is, the temperature that unmolten mantle would have if it were taken adiabatically to a pressure of 1 bar). Our data indicate that older volcanoes are produced from a hotter mantle than



**Figure 4 | Lithosphere thickness as a function of mantle potential temperature estimated from pressure and degree of partial melting.** Heat flow contours (milliwatts per square metre) are calculated from the lithosphere thickness and the melting temperature assuming a conductivity of  $3.5 \text{ W m}^{-1} \text{ K}^{-1}$ .  $1\sigma$  error bars and ellipses are as calculated in Fig. 3.

younger ones, with a total spread of  $\sim 80$  K (Fig. 3). This cooling is associated with thickening of the lithosphere, from 100 to  $\sim 150$  km (Fig. 4), implying a heat flow decrease over time from 38 to  $26 \text{ mW m}^{-2}$  (Fig. 4). These values are in agreement with those derived independently from the thickness of the elastic lithosphere required to support major topographic features:  $20$ – $25 \text{ mW m}^{-2}$  for the young Tharsis Montes,  $\sim 30 \text{ mW m}^{-2}$  for Alba Patera and Elysium Mons, and  $30$ – $40 \text{ mW m}^{-2}$  for the Hesperian volcanism<sup>22,23</sup>. Furthermore, the geochemically derived decrease of  $12 \text{ mW m}^{-2}$  associated with a



**Figure 5 | Mantle temperature variation as a function of the ages of Hesperian volcanism ( $t_1$ ) and Amazonian volcanism ( $t_2$ ).** The inset illustrates this calculation for the variation of heat flow as a function of time. In more detail, temperatures are calculated from integration between  $t_1$  and  $t_2$  of the difference between heat production in the mantle (solid line) and the inferred heat flow decrease from  $36$  to  $28 \text{ mW m}^{-2}$  (dotted line), divided by the heat capacity of the planet. Two possibilities for the distribution of heat-producing elements in the mantle are considered. For the first, values for heat-producing elements are those

of DW85. For the second, a depleted mantle composition has been calculated assuming a value of  $5.17 \times 10^{23} \text{ kg}$  for the mantle mass<sup>29</sup>, a crustal mass of  $1.74 \times 10^{22} \text{ kg}$  (corresponding to an average crustal thickness of  $46 \text{ km}$  derived from MOLA data<sup>30</sup>), average potassium and thorium abundances from GRS data<sup>2</sup>, and a Th/U ratio of  $3.6$  (ref. 15). Calculated temperature variations (degrees Celsius) are shown as grey contours for the case of depleted mantle. Acceptable solutions, corresponding to a temperature decrease of  $80 \pm 20^\circ \text{C}$ , have been highlighted in blue for the depleted mantle and in green for the primitive mantle.



drop of 80 K in mantle temperature is compatible with numerical simulations of the thermal evolution of Mars<sup>3</sup>.

The timescale of cooling is an additional point of interest and may be constrained by integration over time of the difference between internal heat production and the geochemically derived surface heat loss. Assuming reasonable values for heat-producing elements in the mantle<sup>15</sup> and a variation in heat flow from 38 mW m<sup>-2</sup> at  $t_1$  (the Hesperian period) to 26 mW m<sup>-2</sup> at  $t_2$  (the Amazonian period), a temperature variation may be calculated (Fig. 5). Combinations of  $t_1$  and  $t_2$  consistent with a temperature decrease of  $80 \pm 20$  K imply an age for Amazonian volcanism significantly younger than 2 Gyr (Fig. 5), a result consistent with crater counts in the Tharsis region<sup>24–26</sup>.

For the interval from the Hesperian and Amazonian periods, we calculate a mantle cooling rate of 30–40 K Gyr<sup>-1</sup>, associated with a thickening of the lithosphere of 17–25 km Gyr<sup>-1</sup>. This cooling rate is significantly lower than the 70–100 K Gyr<sup>-1</sup> proposed for the Earth over the last 3.5 Gyr (ref. 4). The higher cooling rate for the Earth is related to a significant imbalance between heat production in the mantle and surface heat loss. The ratio of these two quantities is known as the Urey ratio; for the Earth this ratio is  $\sim 0.33$  (refs 5, 6) whereas the value we derive for Mars is  $\sim 0.6$ – $0.7$ . This range is close to the value of  $\sim 0.75$  predicted from thermal evolution models based on Rayleigh–Nusselt scaling laws<sup>5,6</sup>. This comparison suggests that the low terrestrial value of the Urey ratio is the result of some specific feature of the Earth, such as a prolonged period of plate tectonics or the possible existence of a deep magma ocean that concentrates radiogenic elements<sup>27</sup>.

Finally, we note that the mantle temperature of Mars is inferred to be barely lower than that of the Earth<sup>4</sup>; thus, temperature is not responsible for the lack of present volcanic activity. The decrease in the surface expression of magmatism is therefore related to thickening of the lithosphere, which terminates melting at greater depth, reducing melt volumes and the possibility of their ascending to the surface. The same may be true for other apparently extinct planetary bodies, which may remain internally hot and dynamically active. Overall, a self-consistent picture of the thermal evolution of Mars emerges, much simpler than that of the Earth, and provides a useful reference for planetary evolution.

## METHODS SUMMARY

The composition in terms of major constituents (SiO<sub>2</sub>, FeO, Al<sub>2</sub>O<sub>3</sub>, MgO, CaO, K<sub>2</sub>O, Na<sub>2</sub>O, TiO<sub>2</sub>, Cr<sub>2</sub>O<sub>3</sub>, MnO and P<sub>2</sub>O<sub>5</sub>) of liquids produced by melting of the DW85 mantle has been determined using the thermodynamic calculator pMELTS<sup>20</sup> for degrees of partial melting,  $F$ , in the range of 1–20% and pressure of melting,  $P$ , in the range of 1–3 GPa. The pMELTS software has been developed primarily for terrestrial systems, but comparison with experimental data<sup>28</sup> allows quantitative application to more iron-rich systems relevant to Mars<sup>10</sup>. We then determined optimal values of  $P$  and  $F$  for each volcanic province by least-squares adjustment using pMELTS calculations.

**Full Methods** and any associated references are available in the online version of the paper at [www.nature.com/nature](http://www.nature.com/nature).

**Received 4 August 2010; accepted 1 February 2011.**

**Published online 6 April 2011.**

1. Bibring, J.-P. *et al.* Global mineralogical and aqueous Mars history derived from OMEGA/Mars Express data. *Science* **312**, 400–404 (2006).
2. Boynton, W. V. *et al.* Concentration of H, Si, Cl, K, Fe, and Th in the low- and mid-latitude regions of Mars. *J. Geophys. Res.* **112**, E12S99 (2007).
3. Hauck, S. A. & Phillips, R. J. Thermal and crustal evolution of Mars. *J. Geophys. Res.* **107**, E75052 (2002).
4. Herzberg, C., Condie, K. & Korenaga, J. Thermal history of the Earth and its petrological expression. *Earth Planet. Sci. Lett.* **292**, 79–88 (2010).

5. Korenaga, J. Urey ratio and the structure and evolution of Earth's mantle. *Rev. Geophys.* **46**, RG2007 (2008).
6. Jaupart, C., Labrosse, S. & Mareschal, J.-C. in *Treatise on Geophysics* Vol. 7: *Mantle Dynamics* (ed. Bercowski, D.) 253–303 (Elsevier, 2007).
7. Nisbet, E. G., Cheadle, M. J., Arndt, N. T. & Bickle, M. J. Constraining the potential temperature of the Archean mantle: a review of evidence from komatiites. *Lithos* **30**, 291–307 (1993).
8. Karunatilake, S. *et al.* Chemically striking regions on Mars and Stealth revisited. *J. Geophys. Res.* **114**, E12001 (2009).
9. Gasnault, O. *et al.* Quantitative geochemical mapping of martian elemental provinces. *Icarus* **207**, 226–247 (2010).
10. El Maarry, M. R. *et al.* Gamma-ray constraints on the chemical composition of the martian surface in the Tharsis region: a signature of partial melting of the mantle? *J. Volc. Geotherm. Res.* **185**, 116–122 (2009).
11. Taylor, G. J. *et al.* Variations in K/Th on Mars. *J. Geophys. Res.* **111**, E03S06 (2006).
12. Filiberto, J. & Treiman, A. Chlorine-rich, water-poor Martian magmas. *Geochim. Cosmochim. Acta* **73** (suppl.), A376 (2009).
13. Haskin, L. A. *et al.* Water alteration and rocks and soils on Mars at the Spirit rover site in Gusev crater. *Nature* **436**, 66–69 (2005).
14. Bogard, D. D. & Johnson, P. Martian gases in an Antarctic meteorite? *Science* **221**, 651–654 (1983).
15. Dreibus, G. & Wänke, H. Mars, a volatile-rich planet. *Meteoritics* **20**, 367–381 (1985).
16. Filiberto, J., Nekvasil, H. & Lindsley, D. H. The Mars/Earth dichotomy in Mg/Si and Al/Si ratios: is it real? *Am. Mineral.* **91**, 471–474 (2006).
17. Taylor, G. J. *et al.* Bulk composition and early differentiation of Mars. *J. Geophys. Res.* **111**, E03S10 (2006).
18. Monders, A., Medard, E. & Groove, T. L. Phase equilibrium investigations of the Adirondack class basalts from the Gusev plains, Gusev crater, Mars. *Meteor. Planet. Sci.* **42**, 131–148 (2007).
19. McSween, H. Y. *et al.* Characterization and petrologic interpretation of olivine-rich basalts at Gusev Crater, Mars. *J. Geophys. Res.* **111**, E02S10 (2006).
20. Giorso, M. S., Hirschmann, M. M., Reiners, P. W. & Kress, V. C. III. The pMELTS: a revision of MELTS for improved calculation of phase relations and major element partitioning related to partial melting of the mantle 3 GPa. *Geochim. Geophys. Geosyst.* **3**, 1030 (2002).
21. McKenzie, D. & Bickle, M. J. The volume and composition of melt generated by extension of the lithosphere. *J. Petrol.* **29**, 625–679 (1988).
22. McGovern, P. J. *et al.* Localized gravity/topography admittance and correlation spectra on Mars: implications for regional and global evolution. *J. Geophys. Res.* **107**, E125136 (2002).
23. Belleguic, V., Lognonné, P. & Wiczeorek, M. Constraints on the Martian lithosphere from gravity and topography data. *J. Geophys. Res.* **110**, E11005 (2005).
24. Neukum, G. *et al.* Recent and episodic volcanic and glacial activity on Mars revealed by the High Resolution Stereo Camera. *Nature* **432**, 971–979 (2004).
25. Hauber, E., Bleacher, J., Gwinner, K., Williams, D. & Greeley, R. The topography of low shields and associated landforms of plains volcanism in the Tharsis region of Mars. *J. Volcan. Geotherm. Res.* **185**, 1–2, 69–95 (2009).
26. Mangold, N. *et al.* Mineralogy of recent volcanic plains in the Tharsis region, Mars, and implications for platy-ridged flow composition. *Earth Planet. Sci. Lett.* **194**, 3–4, 440–450 (2009).
27. Labrosse, S., Hernlund, J. W. & Coltice, N. A crystallizing dense magma ocean at the base of the Earth's mantle. *Nature* **450**, 866–869 (2007).
28. Bertka, C. M. & Holloway, J. R. Anhydrous partial melting of an iron-rich mantle II: primary melt compositions at 15 kbar. *Contrib. Mineral. Petrol.* **115**, 323–338 (1994).
29. Mocquet, A., Rosenblatt, P., Dehant, V. & Verhoeven, O. The deep interior of Venus, Mars, and the Earth: a brief review and the need for planetary surface-based measurements. *Planet. Space Sci.* (in the press).
30. Zuber, M. T. The crust and mantle of Mars. *Nature* **412**, 220–227 (2001).

**Supplementary Information** is linked to the online version of the paper at [www.nature.com/nature](http://www.nature.com/nature).

**Acknowledgements** H. Y. McSween Jr is thanked for his formal review. This work was financially supported by the Programme National de Planétologie of INSU-CNRS.

**Author Contributions** D.B. and M.J.T. performed calculations of melt–solid equilibria, M.M. and D.B. performed calculations of thermal evolution and O.G. provided expertise concerning the GRS data. All authors participated in the writing of the paper.

**Author Information** Reprints and permissions information is available at [www.nature.com/reprints](http://www.nature.com/reprints). The authors declare no competing financial interests. Readers are welcome to comment on the online version of this article at [www.nature.com/nature](http://www.nature.com/nature). Correspondence and requests for materials should be addressed to D.B. ([baratoux@ntp.obs-mip.fr](mailto:baratoux@ntp.obs-mip.fr)).

## METHODS

**Pressure and degree of partial melting from SiO<sub>2</sub>, FeO and Th.** SiO<sub>2</sub> and FeO concentrations of primary melts formed in the martian mantle are functions of the degree of partial melting,  $F$ , and the pressure,  $P$ , at which the liquid was last in equilibrium with that mantle. pMELTS<sup>20</sup> simulations have been performed to model partial melting of the composition proposed by Dreibus and Wänke<sup>15</sup> (hereinafter DW85) for pressures in the range of 1–3 GPa (in steps of 0.1 GPa) and degrees of partial melting from 1 to 50%. In this way, variations of each oxide component (for example FeO) can be mapped in  $P$ – $F$  space. With these reference grids, the values of FeO and SiO<sub>2</sub> determined from the GRS data may be converted to optimal values of  $P$  and  $F$  through a least-squares adjustment. Although the principle is simple, the following illustrates to what extent this inverse problem is well constrained. Considering a single oxide (SiO<sub>2</sub> or FeO), a family of solutions is obtained and can be represented as a curve in  $P$ – $F$  space. The optimum  $P$  and  $F$  values are found at the intersection of the two curves (see Supplementary Information for the representation of such curves for three Hesperian and three Amazonian volcanic provinces). FeO and SiO<sub>2</sub> contents in the melt are both sensitive to the depth of partial melting and  $P$  seems to be well constrained. A larger range of values of  $F$  may generally explain the SiO<sub>2</sub> and FeO abundances within the error. Using the relationship among the ratio of thorium abundances in the mantle, that in the volcanic rocks and the degree of partial melting, the scale for the degree of partial melting can be converted into a scale indicating the thorium abundance in the mantle source. This exercise shows, for the case of Hesperian volcanism, that the range of  $F$  values determined from SiO<sub>2</sub> and FeO is compatible with the thorium abundance of the primitive martian mantle of DW85. However, for Amazonian volcanic provinces, this exercise implies that the mantle source may be depleted in thorium relative to DW85, although alternative explanations are considered below.

**The influence of crystallization/assimilation on the primary melt assumption.**

Assuming that the surfaces described here consist of primary mantle melts leads to a self-consistent geochemical picture, at least for Hesperian volcanism. Despite this success, we have also explored the effects of fractional crystallization and assimilation of cumulates occurring before emplacement at the surface. The aim of this exercise is to assess to what extent the results of our modelling are sensitive to these processes and to determine whether the apparent variation in the thorium content of the Amazonian mantle may be explained in an alternative way. Olivine being the liquidus phase, determining the composition of the liquid as a function of the degree of fractional crystallization requires the calculation of the composition of olivine in equilibrium with the liquid, given by its Mg#. This ratio is calculated from magnesium–iron partition coefficients<sup>31</sup>. A third dimension has been added to the  $P$ – $F$  grid discussed above, such that for a given GRS datum, values of  $P$  and  $F$  may be calculated for a given degree of fractional crystallization. The results of this exercise are also illustrated in Supplementary Information. For these calculations, the degree of partial melting has been fixed by the thorium content, assuming that the mantle has the thorium abundance proposed by DW85. We find that for Hesperian volcanism, geochemically acceptable values of pressure vary by less than 20% for the studied

range of fractional crystallizations. For Amazonian volcanism, we find that fractional crystallization is not capable of accounting for the discrepancy between pressures determined from SiO<sub>2</sub> and FeO.

Similarly, the effect of assimilation of cumulates has been explored, assuming dissolution of olivine. This calculation is essentially the same as that for fractional crystallization, except that olivine is added rather than subtracted from the primary liquid. Pressure is again calculated from SiO<sub>2</sub> and FeO abundances, assuming the mantle thorium value of DW85 (see Supplementary Information for an illustration of the results). For Hesperian volcanism, consideration of assimilation generally leads to increasing discrepancies between the respective pressures of melting derived from SiO<sub>2</sub> and FeO. For the case of young volcanism, assimilation leads to a convergence of the pressures derived from SiO<sub>2</sub> and FeO, but values of the degree of assimilation required for overlap are considered too high to be a plausible alternative to variation in the thorium content of the mantle.

**The influence of variable Mg# and/or Th of the mantle source.** Given that the effects of fractional crystallization and assimilation of cumulates are not able to reconcile the geochemical data with a single  $P$ – $F$  solution for young volcanism, a chemical modified mantle source seems inevitable for this group. One possibility is variations in Mg#. This has been explored by calculating  $P$ – $F$  grids of SiO<sub>2</sub> and FeO for mantle compositions identical to that of DW85, but with modification of Mg/Fe. In an analogous way to the case of fractional crystallization, these grids have been used to derive values of pressure from SiO<sub>2</sub> and FeO, assuming the thorium abundance of DW85 to constrain  $F$ . As illustrated in Supplementary Information, alternative values of Mg# are not compatible with GRS data for Hesperian volcanism, providing further independent evidence in favour of the idea that the martian mantle is significantly richer in iron than is that of the Earth. For the younger volcanism, we note that a mantle with a Mg# greater than that proposed by DW85 is capable of providing solutions consistent with the SiO<sub>2</sub>, FeO and thorium concentrations of surface lavas. However, given that the older Hesperian lavas are directly consistent with the DW85 composition, it would seem reasonable to assume that the Amazonian mantle is derived from that composition. In this case, episodes of melt extraction will affect both the Mg# and the thorium abundance in ways controlled by the partitioning behaviour of these elements. Two factors influence the variations in the Mg# and the thorium content of the residual mantle following melt extraction. The first is the degree of melting and the second is the fraction of the mantle that experiences this melting. Consideration of possible trajectories in Mg#–Th space (Supplementary Information) shows that whereas thorium content may be significantly affected even by small degrees of melting, variations in Mg# are relatively limited. Therefore, a possible evolution of the Mg# of the martian mantle is not ruled out, but the depletion in thorium is considered to be the dominant factor affecting the mantle source of young volcanism.

- Toplis, M. J. The thermodynamics of iron and magnesium partitioning between olivine and liquid: criteria for assessing and predicting equilibrium in natural and experimental systems. *Contrib. Mineral. Petrol.* **149**, 22–39 (2005).

## CORRIGENDUM

doi:10.1038/nature10220

## Thermal history of Mars inferred from orbital geochemistry of volcanic provinces

David Baratoux, Michael J. Toplis, Marc Monnereau & Olivier Gasnault

*Nature* **472**, 338–341 (2011)

An error in the calculation of heat flow contours drawn in Fig. 4 of our Letter was drawn to our attention by J. Ruiz (Universidad Complutense de Madrid). A surface temperature of 200 °C was mistakenly used instead of the correct temperature of 220 K. With the stated value of thermal conductivity ( $3.5 \text{ W m}^{-1} \text{ K}^{-1}$ ), heat flows are about 20% higher than those originally shown. The knock-on effect of this fact is that the calculated Urey ratio is 20% lower than stated, but still comfortably above the terrestrial value. On the other hand, we note that the value of thermal conductivity relevant to planetary mantles is not well constrained, with preferred values covering the range 2.5 to  $4 \text{ W m}^{-1} \text{ K}^{-1}$  (ref. 1). A value of  $3.0 \text{ W m}^{-1} \text{ K}^{-1}$  and the correct surface temperature leads to calculated heat flow very similar to those shown in our original Fig. 4. A corrected version of Fig. 4, using a surface temperature of 220 K and a conductivity of  $3.5 \text{ W m}^{-1} \text{ K}^{-1}$ , is shown below.

Figure 4 has also been corrected in the original HTML and PDF.

1. Breuer, D. & Moore, W. B. in *Treatise of Geophysics* Vol. 10, 299–341 (Elsevier, 2011).

